

Synchronized EEG-compatible cognitive experiment using LSL

Introduction

This report describes the design and implementation of an Eriksen Flanker Task using PsychoPy Builder. The Flanker Task is a well-established cognitive paradigm used to study selective attention and cognitive control.

In this task, participants are required to identify a central target stimulus while ignoring surrounding distracting stimuli (flankers). The critical manipulation is whether the flankers are:

- **Congruent** (same as the target)
- **Incongruent** (different from the target)

The present implementation uses letter stimuli (*H* and *A*) arranged in five-character strings. The central character is the target, and the surrounding four characters act as distractors.

Examples of the experimental conditions are shown below:

STIMULUS	CORRECT ANSWER	CONDITION
HHHHH	f	congruent
AAAAA	j	congruent
HHAHH	j	incongruent
AAHAA	f	incongruent

Table 1. Conditions.csv

This paradigm is particularly relevant in the context of EEG and Brain-Computer Interface (BCI) research. Incongruent trials are known to elicit specific event-related potentials (ERPs), such as the N200 and P300 components, associated with conflict monitoring and cognitive control.

Hypothesis

It is expected that:

- Reaction times will be longer in incongruent trials
- Accuracy will be lower in incongruent trials

This effect, known as the Flanker effect, reflects increased cognitive interference when target and distractors are incongruent.

Experimental Structure

The experiment was built in PsychoPy Builder and follows a structured sequence of loops and routines:

1. Instructions routine:

This routine presents a full-screen explanation of the task, including:

- The goal of the experiment
- The response keys (F and J)
- Instructions to focus on the central letter

Participants are required to press any key to begin the experiment.



Image 1. Instructions routine

2. Practice loop:

A practice block is included to allow participants to become familiar with the task before the main experiment.

- Repetitions: 1 per condition
- Total trials: 4
- Order: randomized
- Conditions: conditions.csv

This helps reduce learning effects and ensures that participants understand the task requirements.

Each practice trial consists of the following routines:

- Fixation routine** - A fixation cross (+) appears at the center of the screen for a variable duration between 700 and 1000 ms, its purpose is to direct the participant's attention and reduce anticipatory responses.
- Stimulus routine** - A five-letter string is presented on the screen. Participants must respond according to the central letter. The response window remains

active until a key press is detected or a maximum duration of 1500 ms is reached. It measures reaction time and accuracy.

- c) **Feedback routine** - After each response to reinforce task understanding and improve performance, feedback is displayed with a duration of 600 ms:
 - “Correct” for correct responses
 - “Error” for incorrect responses
- d) **Between trials routine** - To separate trials and avoid overlap of cognitive processes a blank screen is presented between trials with a duration of 600 ms.

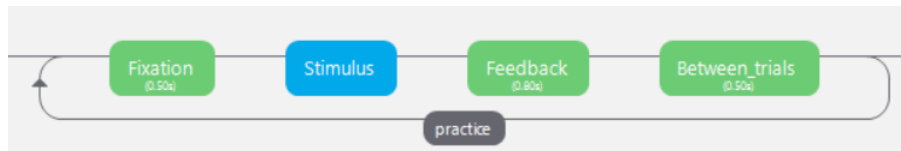


Image 2. Practice loop

3. Start_trials routine - To clearly separate practice from experimental trials. After the practice loop, a transition screen informs the participant that the main experiment is about to begin.

Participants must press any key to continue.

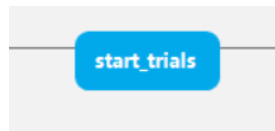


Image 3. Start_trials routine

4. Trials loop

This loop contains the main experimental trials.

- Repetitions: 15 per condition
- Total trials: 60
- Order: randomized
- Conditions: conditions.csv

The structure of each trial is identical to the practice loop but it is not a practice, it is collecting behavioral data for the analysis

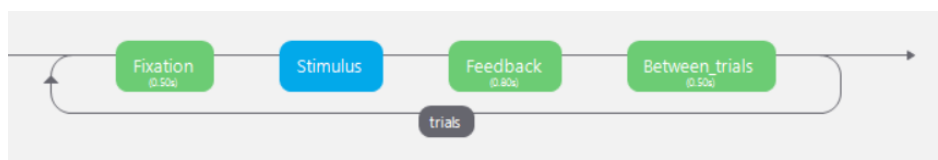


Image 4. Trials loop

Trial Timing

Each trial consists of four sequential phases:

Phase	Duration	Purpose
Fixation cross	700 - 1000 ms	Prepares attentional focus
Stimulus display	Until response (max 1500 ms)	Response window
Feedback display	600 ms	Performance feedback
Interval between trials	500 ms	Temporal separation

Table 2. Trail timing distribution with functional justification.

These parameters are consistent with standard Flanker Task protocols reported in the literature.

Data Collection

The experiment records the following variables:

- Reaction time (RT)
- Accuracy (correct vs incorrect responses)
- Condition (congruent vs incongruent)

Trials are presented in a randomized order to prevent sequence effects.

Additionally, synchronized LSL event markers can be recorded together with EEG signals for neurophysiological analysis.

LSL Integration

The experiment was integrated with Lab Streaming Layer (LSL) to enable real-time synchronization between stimulus presentation and EEG acquisition.

LSL was used to transmit event markers from PsychoPy during the execution of the task. Each experimental event generates a marker that can later be synchronized with EEG recordings obtained from the Unicorn Hybrid Black device.

The synchronization pipeline consists of:

1. PsychoPy stimulus presentation
2. LSL event markers
3. LabRecorder synchronization
4. EEG + event streams stored in .xdf format

This architecture allows precise temporal alignment between cognitive events and neural activity, which is essential for ERP analysis.

Event Markers

Event markers are transmitted through the LSL stream whenever a stimulus is presented.

Marker	Description
Congruent	Congruent stimulus presentation
Incongruent	Incongruent stimulus presentation
Correct	Correct participant response
Incorrect	Incorrect participant response

Table 3. Event markers description.

These markers are recorded simultaneously with EEG signals using LabRecorder, enabling offline synchronization and analysis.

EEG Compatibility

The task was specifically designed to be compatible with EEG acquisition systems through LSL communication.

The Unicorn Hybrid Black EEG device streams EEG signals through its own LSL outlet, while PsychoPy simultaneously transmits event markers. Both streams are recorded together using LabRecorder in a synchronized .xdf file.

This setup enables the extraction of event-related potentials (ERPs) associated with cognitive conflict during incongruent trials.

Potential EEG Analysis

The synchronized EEG and event-marker recordings obtained with this paradigm can be used to analyze event-related potentials (ERPs) associated with attentional control and cognitive conflict.

In particular, incongruent trials are expected to produce enhanced N200 responses related to conflict monitoring, as well as modulations in the P300 component associated with attentional processing.

The resulting .xdf recordings can be processed offline using MATLAB or Python toolboxes compatible with LSL data structures.

References

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